

High Temperature Alert System Using ESP32

Components Used:

- ESP32 DevKit
- DHT22 Temperature Sensor
- Red LED
- 220Ω Resistor
- Buzzer

Software Used:

- Wokwi Simulator
- Arduino IDE

Sensor Used:

- DHT22 Temperature Sensor

Working:

The DHT22 sensor continuously measures the surrounding temperature and sends the data to the ESP32. If the temperature goes above the set limit (e.g., 35°C), the ESP32 turns ON the red LED and buzzer to alert the user. When the temperature comes back below the limit, both the LED and buzzer turn OFF.

Input: Temperature from DHT22 sensor.

Output: Red LED and buzzer alert.

Applications:

- Home Safety
- Server Rooms
- Industries
- Laboratories

Conclusion:

This project monitors temperature in real time using ESP32. It provides an instant alert through

an LED and buzzer whenever the temperature exceeds the predefined limit, helping prevent overheating.

Project Link

<https://wokwi.com/projects/469996857454030849>